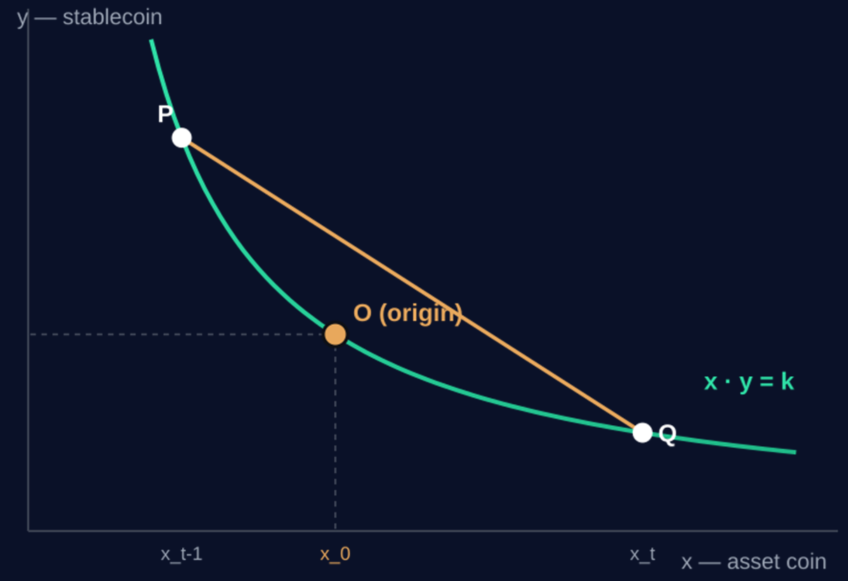
BUILT ON GIWA · GASOK 2026

GainX

— the origin-crossing AMM



The AMM that turns impermanent loss into impermanent gain.

ORIGIN LABS Impermanent-Loss Mitigation Protocol for DEXs

THE PROBLEM

Impermanent loss is the silent tax on liquidity

Every AMM runs on liquidity providers (LPs). But on a constant-product AMM, every price move lets arbitrageurs rebalance the pool — the LP is left holding more of the falling asset and less of the rising one.

The result: LPs often end up worth less than if they had simply held — even after fees. It is the #1 reason capital stays on the sidelines.

49.5% of Uniswap v3 LPs ended with negative returns vs. just holding

$260M in impermanent loss vs. $199M in fees, across the 17 largest v3 pools

14/17 of those pools saw impermanent loss exceed all fees earned

Source: Loesch et al., Topaze Blue & Bancor, 2021 (arXiv:2111.09192).

WHY IT PERSISTED

A decade of fixes failed — the math used the wrong price

Today's AMMs: the reference price

Price is read off the pool state alone — Markov Process' Zero-order model: $P_{\text{ref}} = \frac{k}{x^2}$

Under this price, the LP value gain G is mathematically ≤ 0 for every trade

Loss isn't bad luck — it is guaranteed by the pricing model itself.

What's been tried

- Option hedges — costly, needs external markets
- Stable-only pairs — avoids IL by avoiding volatility
- Liquidity-mining rewards — subsidizes, never removes the loss
- Concentrated liquidity (v3) — actually amplified IL for most LPs

If the price model guarantees loss, no incentive can fully cancel it. The fix has to start with the math.

OUR BREAKTHROUGH

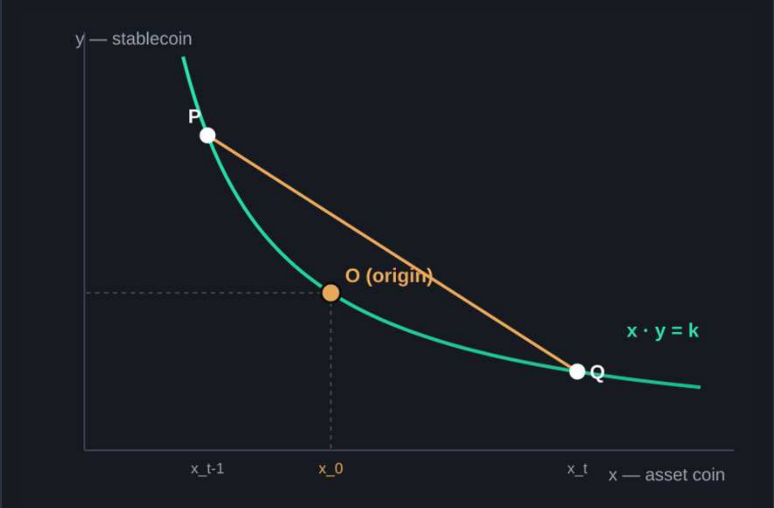
We price by what actually happened

Actual price. We price each swap by what was really exchanged — the realized ratio – actual price (a Markov Process' First-order, Path-aware model), not the pool's instantaneous quote.

$$P_{act} = -\frac{\Delta y}{\Delta x}$$

Origin crossing (proven). When a trade moves the pool across its starting point — $x_{t-1} < x_0 < x_t$ (or the reverse) — the LP value gain is strictly positive. **Impermanent gain.**

Lee, G. M., & Kim, H. J. (2025). "Impermanent Loss Mitigation for Decentralized Exchanges through Optimization". International Journal of Industrial Engineering: Theory, Applications and Practice, 31(6).



Origin O lies between the trade's endpoints P→Q

⇒ Value gain G > 0.

HOW IT WORKS

Batch and reorder the block to manufacture origin crossings

1 · Capture

Orders for GainX arrive within each ~1-second block.

→

2 · Batch

Aggregate same-direction orders into one trade at a uniform clearing price — lower gas.

→

3 · Reorder

Sequence the batches so trades cross the origin → LP value gain, with slippage minimized and fairly shared.

Same trades. Same average price. ***Re-sequenced so the pool and its LPs come out ahead — MEV redirected to liquidity, not extracted against users.***

PROOF

Benchmark in our research: −95% impermanent loss

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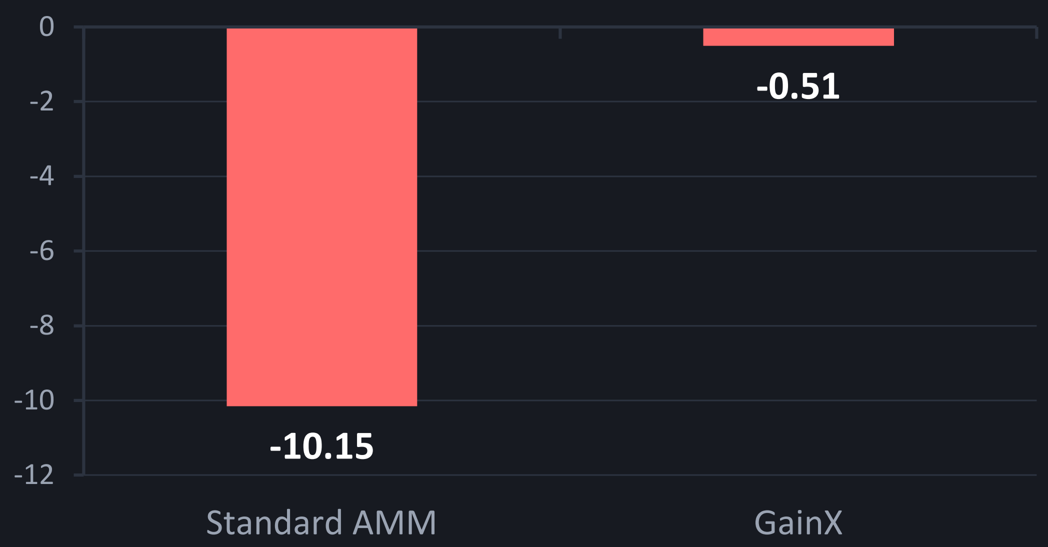


Net cumulative LP value gain over the identical 17-trade flow. Source: Lee & Kim (2024), Tables 1–2.

17 → 8 trades collapsed into batches — fewer on-chain txns, lower gas

7 of 8 batches avoid impermanent loss (4 gain, 3 break-even)

+1.12 peak cumulative LP value gain during the run

−0.5 tail an end-of-block boundary effect; continuous flow keeps producing gain

WHY GIWA

GIWA is where this protocol becomes practical

OP Stack sequencer

Batching & reordering are programmable at DEX front-end and smart contract. The OP Stack sequencer provides it natively — impossible to coordinate on a generic L1.

1s blocks + Flashblocks

Fast, dense blocks supply enough concurrent order flow to batch and reorder pools every single block.

Powered by Upbit

Direct rails to real liquidity and users — pools can be bootstrapped with depth from day one.

Open-source & EVM

GainX can propose an open protocol. GIWA shares that exact philosophy and Solidity tooling.

ECOSYSTEM FIT

Native to GIWA Wallet, Dojang, and UP.ID

GIWA Wallet

One-tap LP onboarding inside the wallet: deposit, watch value gain accrue, and withdraw — fully in-app. Built for the onboarding criterion.

Dojang

Verified-Address attestations gate LP and relayer roles → Sybil-resistant, auditable, and fair distribution of captured value.

UP.ID

Human-readable LP and pool identities (alice.up.id) make GainX markets transparent and trusted by default.

Every criterion the GASOK program scores — chain fit, originality, feasibility, market, team, wallet — maps to a piece we already have.

MARKET

Fix IL, and you unlock the liquidity flywheel

LPs protected
→ abundant liquidity → lower slippage
→ more traders & volume
→ more fees → more LPs

Liquidity is the one thing every L2 fights for. GainX makes GIWA the most LP-friendly chain in DeFi — a structural reason to bring capital here first.

$500B+	monthly DEX spot volume in 2025 (DefiLlama)
~half	of v3 LPs lost money to IL — a still-unsolved, universal pain
1st	AMM built on a proven impermanent-gain condition

Sources: DefiLlama (DEX spot volume, 2025); Loesch et al., 2021 (LP returns). Full citations on the final slide.

ROADMAP

From proven theory to GIWA mainnet

Phase 0 — Done

Origin-crossing proven; batching + reordering formulated as a combinatorial optimization problem. Peer-reviewed, 2025.

Phase 1 — GASOK · M1-2

CPMM pool + batch/reorder optimizer (Ultra-speed greedy/two-pointer heuristics)) live on GIWA testnet, with a GIWA Wallet deposit flow.

Phase 2 — M3-4

Dojang & UP.ID integration; sequencer ordering module; third-party security audit.

Phase 3 — M5-6

Mainnet pools (ETH / stable first) + open-source SDK and protocol spec for the whole GIWA ecosystem.

TEAM

Origin Labs — the researchers who proved it

We are Ph.D. engineers in operations-research and computer science paired with DeFi and smart-contract engineers and developers. The origin-crossing result and the batching/reordering optimization are our own published work — not a borrowed idea. We now build it on-chain.

Core competency: turning a hard combinatorial problem into a working protocol.

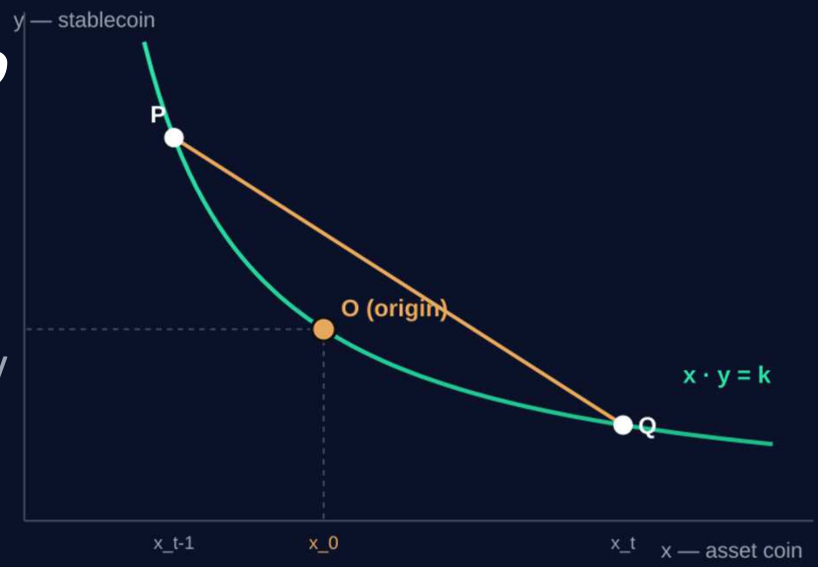
Combinatorial optimization & metaheuristics

AMM design & market microstructure

Solidity / EVM smart-contract engineering

THE ASK

Liquidity that gains, not loses.



Help us make GIWA the first chain where providing liquidity is a winning trade.

With GASOK we'll use: development funding · OP Stack sequencer guidance · Upbit liquidity bootstrap · Dojang / UP.ID integration · a security audit.

ORIGIN LABS · GainX

GainX Architecture: Balancing LPs and Traders

The Challenge: Batching creates a uniform clearing price, leading to an execution price discrepancy for chronological traders.

The Solution: GainX introduces **Slippage Smoothing**. The mathematical surplus (Impermanent Gain) generated from origin-crossing trades is partially redirected to rebate early traders.

The Outcome: LPs achieve mathematically proven gains instead of losses, while traders benefit from lower gas fees and guaranteed fair-slippage boundaries.

SELF-EVALUATION

GainX Self-Score Card & Highlights (1/6)

1. GIWA Chain Suitability

Excellent

The Next "Aerodrome" of GIWA: Ecosystem Native Infrastructure

- **GIWA Wallet:** One-touch in-app LP onboarding—deposit, track real-time Impermanent Gain (IG), and withdraw seamlessly.
- **Dojang Contract:** Sybil-resistant, verified-address attestations gate LP and relayer roles for fair value distribution.
- **UP.ID:** Replaces complex hex addresses with human-readable IDs (alice.up.id) to build transparent, trusted markets.

SELF-EVALUATION

GainX Self-Score Card & Highlights (2/6)

2. Originality

Excellent

Paradigm Shift in AMM Pricing: Peer-Reviewed & Patent-Backed

- **First-Ever IG Implementation:** The first AMM to mathematically prove, verify, and implement the "Origin Crossing" condition to turn IL into structural gain.
- **IP Entry Barrier:** Built on over a decade of proprietary Operations Research, creating a defensible academic and cryptographic moat.

SELF-EVALUATION

GainX Self-Score Card & Highlights (3/6)

3. Feasibility

Excellent

Ready-to-Deploy Architecture: High-Speed Off-Chain Ordering

- **Production-Ready Modules:** Transaction capturing, directional batching, and two-pointer win-swaps are already fully formulated.
- **1-sec Block Execution:** Native integration with OP Stack sequencers and Flashblocks achieves millisecond-level sorting without on-chain gas overhead.

SELF-EVALUATION

GainX Self-Score Card & Highlights (3/6)

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SELF-EVALUATION

GainX Self-Score Card & Highlights (4/6)

4. Marketability

Excellent

Unlocking the Ultimate DeFi Liquidity Flywheel

- **Solves Universal Pain:** Targets the unsolved problem where ~50% of v3 LPs suffer net losses due to Impermanent Loss.
- **The Flywheel:** Protected LPs → Abandant Liquidity → Lower Slippage → Higher Volume → More Fees. GainX makes GIWA the most lucrative, LP-friendly destination in Web3.

SELF-EVALUATION

GainX Self-Score Card & Highlights (4/6)

5. Team Capability

Excellent

A Decade of Academic Depth Meets Web3 Engineering

- **Operations Research/ Fintech Experts:** Led by PhDs with 10+ years of deep research in combinatorial optimization and market microstructure.
- **Code-Proven Execution:** Native expertise in translating complex path-dependent mathematics into gas-optimized EVM Solidity protocols.

GainX Self-Score Card & Highlights (4/6)

6. GIWA Wallet Integration

Excellent

Zero-Friction In-App Compatibility

- **Seamless API Architecture:** Designed to integrate natively with GIWA Wallet backends without complex front-end overhauls.
- **Zero Onboarding Friction:** LPs enjoy their standard user experience while viewing accumulated GYOCHA rewards directly inside the wallet app.

SOURCES & REFERENCES

Every figure, traced to its source

1 **Impermanent-loss statistics — slides 2 & 9**
Loesch, Hindman, Richardson & Welch, “Impermanent Loss in Uniswap v3,” Topaze Blue & Bancor, 2021 · arXiv:2111.09192. 17 largest v3 pools (43% of TVL), 17,000+ wallets: $260.1M IL vs $199.3M fees; 49.5% of LPs negative; only 3 of 17 pools beat IL.

2 **Origin crossing & optimization results — slides 4 & 6**
Lee & Kim, “Impermanent Loss Mitigation for Decentralized Exchanges through Optimization,” Int'l Journal of Industrial Engineering 31(6):1426–1437, 2024 · DOI:10.23055/ijietap.2024.31.6.10691. Slide-6 values computed from Tables 1–2.

3 **DEX market size — slide 9**
DefiLlama, monthly DEX spot volume, 2025 (e.g. ~$506B in Aug 2025; ~$613B in Oct 2025) · defillama.com/dexs.

4 **GIWA chain & ecosystem — slides 1, 4–8**
GIWA official documentation · docs.giwa.io (OP Stack L2, 1-second blocks, Flashblocks, GIWA Wallet, Dojang, UP.ID; Powered by Upbit).